

# Bayesian Optimisation with Replication Selection for Stochastic Simulation Optimisation

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# Introduction

# Stochastic Simulation

# Bayesian Optimisation (BO)

# Elements of BO

## Our Approach

# Heteroscedastic Gaussian process

Observations from the simulation are described using by a Gaussian Process:

$$y \sim \mathcal{GP}(f, \sigma_{\epsilon}^2(x))$$

# Replication and cost aware Acquisition Function



## Simulation Study

# Test Problem